Wireless Security - GNSS lab activity report

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1 METHODS

This project focuses on the analysis of Android GNSS raw measurements and the emulation of a GNSS spoofing attack in a controlled software environment.

The experimental workflow was divided into two main phases: data collection and offline analysis. GNSS measurements were recorded on an Android smartphone using the GNSS Logger application, with measurement log-on enabled. During acquisition, the app recorded raw GNSS data that were later exported to a PC for post-processing. Timed logging was used to keep acquisitions comparable in duration.

The collected logs were processed in MATLAB using the modified version of Google's GPS measurement tools provided for the lab. In particular, the main script ProcessGnssMeasScript.m was used to parse the raw measurements, compute pseudoranges through ProcessGnssMeas(gnssRaw), and estimate the GNSS PVT solution through GpsWlsPvt(gnssMeas, allGpsEph). This step allowed us to inspect the evolution of pseudorange measurements, clock bias effects, satellite geometry, and the final position estimates. The analysis is based on Android raw fields such as receiver time, clock bias, and satellite transmission time, which are fundamental for pseudorange computation. To study spoofing, we used the spoofing options already available in the enhanced MATLAB tool. The attack was emulated by acting directly on the measurements extracted from the real data collection. The results were evaluated by comparing the baseline and spoofed cases in terms of position displacement, variation of pseudorange-related plots, clock behavior, and overall consistency of the PVT output.

2 RESULTS AND DISCUSSION

The following subsections discuss the results obtained from a dataset collected under minimal interference conditions, and compare them with scenarios in which spoofing techniques were applied, as described in Section 1.

2.1 Baseline Data Collection and Spoofing

The quality of the Signal-to-Noise Ratio (SNR) in the undisturbed case can be appreciated in Figure 1, where most satellite signals are well above 30 dB.

2.1.1 Delay Spoofing. The experiment involved spoofing the position by injecting a time delay of 10 μs after 60 s from the beginning of the data acquisition.

Applying a time-based spoofing attack produces a clear effect on the observables: the pseudoranges (whose derivative over time is shown in Figure 2) and the clock bias offset (see Figure 3) exhibit abrupt jumps. This behaviour justifies the high standard deviation found analyzing latitude and longitude acqusitions, respectively

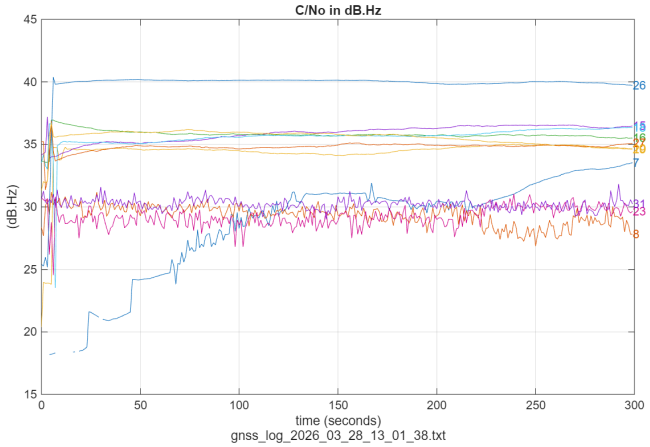


Figure 1: Baseline dataset C/N_0

46m and 10m (see the dispersion on Figure 4). Although the attack is completed successfully, it remains easily detectable.

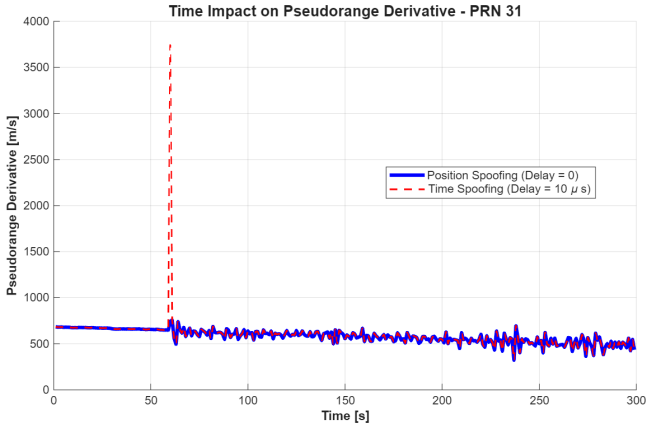


Figure 2: Pseudorange derivative over time

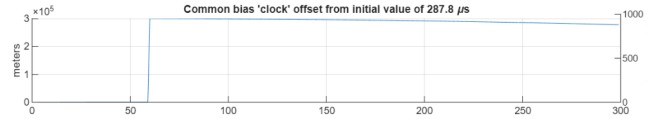


Figure 3: Clock bias offset under delay spoofing

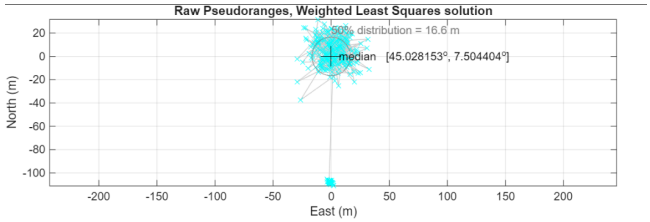


Figure 4: Estimated position under delay spoofing

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2.1.2 Location-Specific Spoofing. By setting specific coordinates through the `spoof.position` array in the script, it is possible to force the receiver toward a predefined location.

This type of spoofing clearly compromises the final estimated position (see Figure 5), despite the presence of the observable discontinuities described in Section 2.1.1.

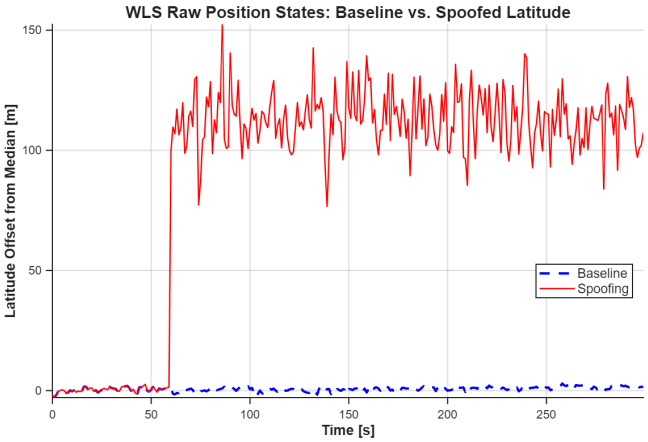


Figure 5: Spoofed position

2.1.3 Enhanced spoofing algorithm. To make the spoofing attack more realistic, it is reasonable to smooth the position shift by distributing it over time, rather than introducing an abrupt change.

To achieve this, we define a variable ramp_duration, which represents the time interval over which the displacement is applied. Then, using a ramp function $\alpha = \frac{t - t_{\text{start}}}{t_{\text{ramp}}}$, this parameter α is progressively added to T_{RX} as the independent variable t evolves, resulting in a gradual, approximately linear shift in the estimated position (see Figure 6).

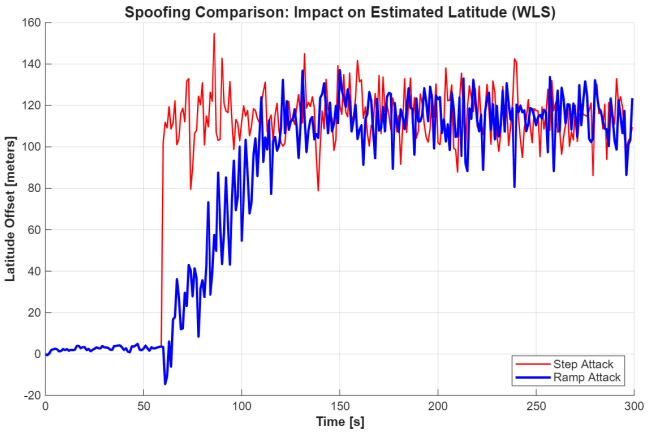


Figure 6: Enhanced spoofing method

2.2 Peculiar Conditions Data Collection

Using the baseline dataset (Section 2.1) as a reference, it is possible to highlight how different operating conditions affect GNSS observables.

2.2.1 Stationary vehicle. Measurements collected inside a stationary vehicle show a clear attenuation of the SNR due to signal obstruction (Figure 7). Diving into the details, the WLS performance degrades, with the error increasing from 1.1 m to 1.6 m, corresponding to a 68% increase. The estimated latitude and longitude in this scenario deviate significantly from the baseline, with errors reaching up to ±20 m from the median. The HDOP also exhibits abnormal behavior, rising above 1.0, which indicates a reduced number of satellites in view (see figure 8).

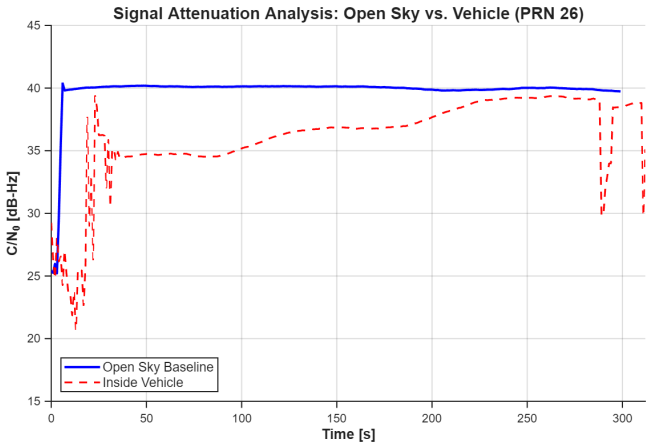


Figure 7: C/N_0 comparison

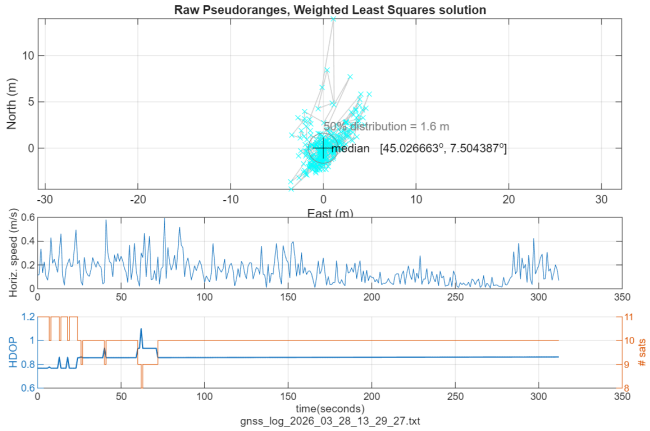


Figure 8: WLS, Horizontal speed and HDOP inside the vehicle

2.2.2 Power saving mode. In the battery save scenario, the solution is still usable, but several signs of degraded measurement continuity appear. The horizontal standard deviation increases from 1.38 m in the baseline to 1.81 m here, which is about a 32% degradation. The latitude standard deviation worsens from 0.972 m to 1.532 m, while longitude remains about the same (0.976 m). So the effect is not a total collapse of positioning, but a clear reduction in stability and robustness. The most important difference is qualitative: compared with the baseline, this dataset shows intermittent tracking behavior, simultaneous C/N0 drops, larger state excursions, and more outliers in the position solution. These are all coherent with receiver duty-cycling or reduced continuous tracking caused by energy-saving operation.

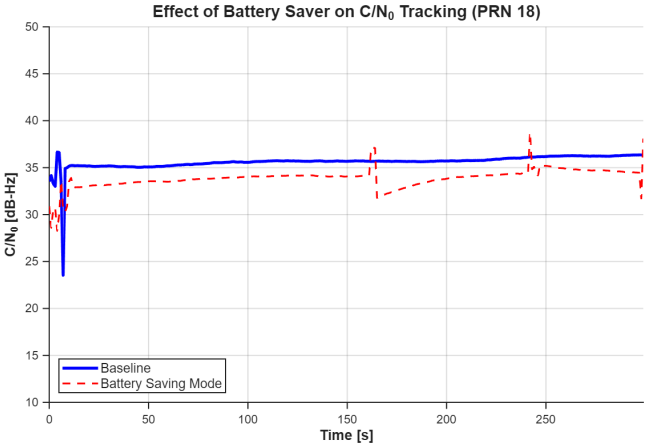


Figure 9: Comparing baseline vs energy saving mode

2.2.3 Enabling hotspot. Compared to the baseline, the hotspot scenario does not significantly degrade horizontal positioning. Horizontal precision remains essentially unchanged, with the standard deviation increasing only slightly, and the WLS 50% circle measuring 1.1m in the baseline versus 1.0m under hotspot conditions. This confirms that the estimated position remains stable despite hotspot activity. Nonetheless, several internal receiver metrics are affected. The receiver clock exhibits significant changes, and the frequency offset no longer follows the clean linear trend observed in the baseline (as shown in Figure 10).

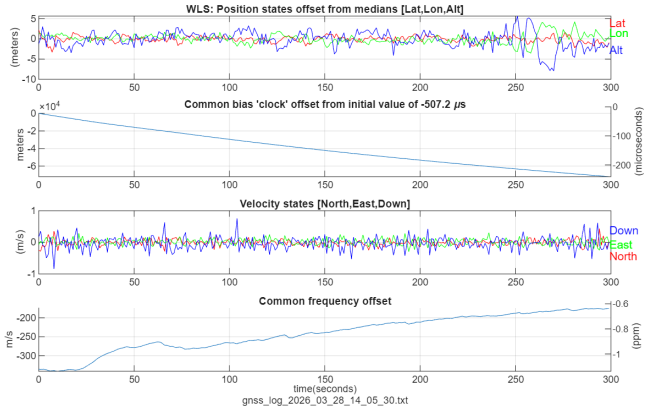


Figure 10: Clock bias and frequency offset with hotspot enabled

The number of tracked satellites decreases from 11 to 10, while HDOP gradually declines rather than remaining flat. Overall, C/N_0 remains high, but some satellites show weaker signals and local disturbances, particularly toward the end of the session.

2.2.4 Moving vehicle. In this experiment, the data were collected during a train journey at the beginning of the X–X route. The rapid variation in position prevents the convergence of a stable final position estimate. Nevertheless, it can be observed that, in some segments, the tracking closely follows the railway path, while in others the trajectory exhibits irregular, piecewise patterns that are not consistent with the typical motion of a train (see Figure 11). This behavior is due to the fact that a final position cannot be obtained by averaging all measurements. Instead, each estimate must be considered as an independent valid solution for the corresponding time instant.

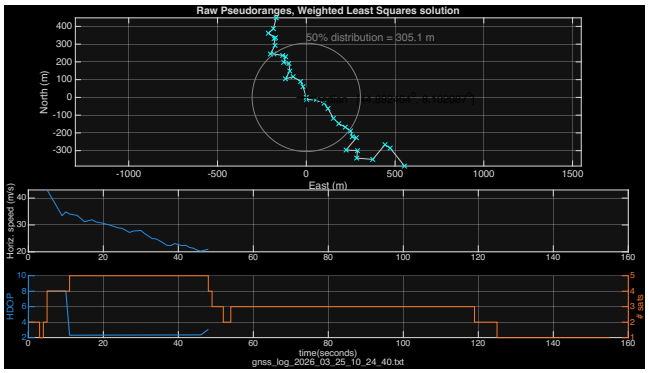


Figure 11: Railway tracking

3 CONCLUSION

This work showed that Android GNSS raw measurements are a powerful tool used for analyzing both spoofing attacks and non-malicious degradations in smartphone positioning. The baseline dataset provided a stable open-sky reference with high C/N_0, regular clock behavior, and a compact WLS solution. Compared against this reference, spoofing attacks produced the most evident effects, including unespected changes in pseudoranges, clock bias anomalies, and large position displacements. A smoother ramp-based spoofing strategy reduced the abruptness of these anomalies, making the attack appear more realistic.

The experiments under peculiar conditions showed that different perturbations affect GNSS in different ways. Measurements inside a stationary vehicle mainly caused signal attenuation, poorer geometry, and larger positioning errors. Power saving mode degraded tracking continuity and produced simultaneous C/N_0 drops and larger state excursions, while hotspot activity left the final horizontal accuracy almost unchanged but perturbed clock-related receiver states and reduced satellite tracking stability. In the moving-vehicle case, the results also showed that a static-style median analysis is no longer appropriate, since each epoch must be treated as an independent solution.

In summary, the study confirms that raw GNSS observables are useful not only for positioning, but also for recognizing abnormal receiver behavior and assessing the detectability of spoofing. Future work could investigate more advanced spoofing strategies and develop automatic detection methods based on joint monitoring of pseudoranges, C/N_0, clock bias, and tracking continuity.

REFERENCES

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